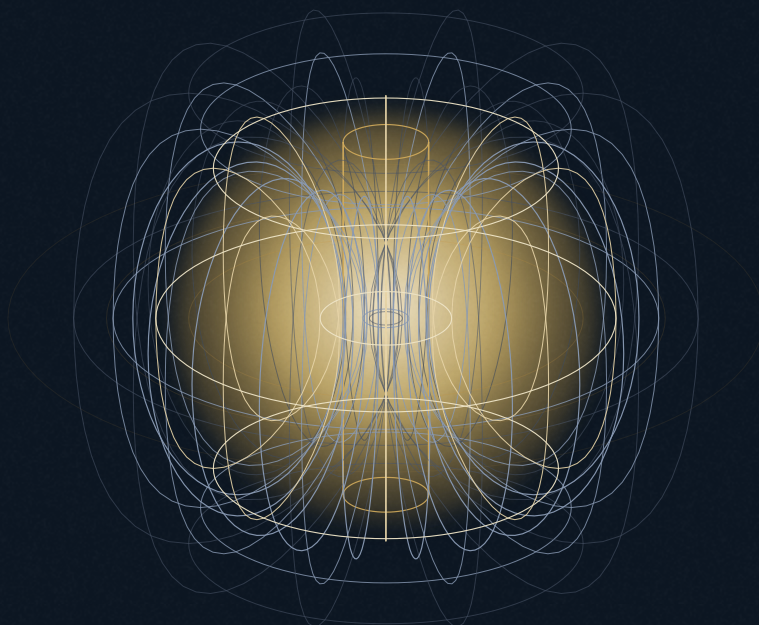




KRONOS FUSION ENERGY



PAPER 2.21 OF THE 2026 KRONOS SET

The Safety Case

Bounded Hazards, Stated Plainly

Fusion's safety advantages are real but often asserted; here they are bounded against the platform's actual hazards.

Mobilizable tritium minimized · no meltdown by construction · low-activation waste target

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THE OFFICIAL RECORD

What this document is, and how to read its numbers

This is a combined editorial. It gathers, in one volume, the two design studies Kronos Fusion Energy released in 2026 — a compact spherical-tokamak tritium/helium-3 breeder and a deuterium–helium-3 tandem-mirror generator — together with the intellectual-property estate, the people, and, in the internal edition, the full commercial case. The scientific text is the same text that appears in the arXiv preprints and the journal submissions; the editorial adds the apparatus a reader needs to hold the whole program at once, and nothing in the physics is altered to fit it.

THE HONESTY STANCE IS THE ASSET

Every headline quantity carries an **evidence class** — DERIVED RECOMPUTED MEASURED REQUIREMENT — and, where a number is a modelled extrapolation rather than a demonstrated value, it is labelled as such and its downside is shown next to it. Several quantities in the underlying corpus moved after first publication; where a value has been withdrawn or restated, the record says so plainly. This is deliberate: a diligence team will find the load-bearing assumptions, so the document surfaces them first.

TITLE	The Kronos Fleet — Hyperion · Aegis · MetroVolt: A Combined Design & Commercialization Study
AUTHOR	Priyanca Ford, Founder & CEO, on behalf of Kronos Fusion Energy
EDITION	2026 Combined Editorial · first issue
COMPANION WORKS	The five-paper arXiv drop — (1) Breeder, (2) Burner, (3) REBCO magnet & tape, (4) Direct Energy Conversion, (5) AI/ML/Quantum control — with matching numbered Zenodo deposits, plus the IOP journal and IAEA-FEC submissions. Patents were filed before the drop.
NAMING	Hyperion = the breeder; Aegis (defense) and MetroVolt (data-center) = one generator in two housings. Internal mode letters (D1, L) do not appear on public surfaces.
STATUS	Conceptual design & simulation study. Nothing herein is a construction commitment.



HOW TO READ THIS VOLUME

The fleet at a glance, and the way its numbers are marked

This volume opens with *Orientation*, presents the five research papers as a bound sequence, and closes with the *Apparatus*. *Foundations* sets out the three general results both machines rest on. *Paper One* is the Hyperion breeder; *Paper Two* the Aegis / MetroVolt generator; *Papers Three, Four and Five* are the enabling science — high-field REBCO magnets and tape, direct energy conversion, and the AI / ML / quantum control stack. A *Digital Twin & 3-D Model* part follows, then the *Environmental* profile. The *Economics* and levelized cost and the *Business Case* and diligence record appear in the internal edition only; the *Simulations* proof record and the Apparatus — patent portfolio, team, limitations, and sources — close both editions.

THE FLEET AT A GLANCE

	HYPERION	AEGIS	METROVOLT
Role	Strategic-isotope foundry	Defense installation power	Campus power
Machine	Spherical tokamak	Tandem mirror	Tandem mirror
Fuel	Deuterium–tritium	Deuterium–helium-3	Deuterium–helium-3
Delivers	Tritium · ³ He · 14 MeV n	Resilient installation power	Firm campus power
Physics bar	Fusion gain only	Closure (gates named)	Closure + lunar ³ He
In the fleet	Breeds the fuel	Proves the generator	Commercial destination

HOW THE NUMBERS ARE MARKED

Every headline quantity carries an evidence class — **DERIVED** from first principles, **RECOMPUTED** from raw data, **MEASURED** in hardware, or a **REQUIREMENT** yet to be met. Financial figures carry a case label and are confined to the internal (confidential) edition. Values that moved after first publication are shown as withdrawn or restated rather than quietly changed. A capability you can evaluate is one whose limits are on the page.

ABSTRACT

The Safety Case — Bounded Hazards, Stated Plainly

Fusion's safety advantages are real but are often asserted rather than bounded. The dominant radiological hazard is the mobilizable tritium inventory, minimized by direct internal recycling rather than a large buffered store; the standing-inventory value that sets the source term is *computed* in the fuel-cycle model, not asserted. There is no fission-product inventory and no chain reaction, so an unmitigated loss of control cannot produce a runaway or a meltdown — the plasma is quenched by loss of any of its several necessary conditions. Structural activation targets a low-level-waste class, and the burner's 5.44% neutron fraction eases power-plant activation.

Companion deposit (code & data, reproducible under CC BY 4.0): DOI [10.5281/zenodo.22132133](https://doi.org/10.5281/zenodo.22132133) (v2 [10.5281/zenodo.21746479](https://doi.org/10.5281/zenodo.21746479)) · Zenodo community *kronos_fusion_energy*.

Explore it live: [interactive 3-D model](#) · [run the physics validator](#) · [companion film series](#) (the deposited solvers, in the browser).

Keywords:

fusion safety

tritium inventory

source term

low-activation steel

passive safety

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NOMENCLATURE

Symbols, units, and the names of things

Q	plasma (fusion) gain, $P_{\text{fus}}/P_{\text{aux}}$
Q_E	engineering gain, net electric out / recirculating in
H_{98}	confinement enhancement over the IPB98(y,2) scaling
Z_{eff}	effective ion charge
c_{ee}	electron–electron bremsstrahlung leading coefficient, 2.120022 (exact)
τ_E	energy confinement time
I_p	plasma current
R_0, a	major radius, minor radius
κ, δ	elongation, triangularity
β_N	normalized plasma beta (Troyon-normalized)
TBR	tritium breeding ratio
DEC	direct energy conversion
CF	plant capacity factor (availability)
FOAK/NOAK/BOAK	first / n th / best of a kind
Hyperion	the ST breeder (internal: Mode D2)
Aegis	the generator, defense/installation housing (internal: Mode M)
MetroVolt	the generator, data-center housing (Mode M)

§ 1

A credible safety case names the hazards

Fusion's safety advantages are real, but they are too often *asserted* rather than bounded. A credible case names the hazards and bounds each one. Fusion's inherent advantages — no fission-product inventory, no criticality, a small driven plasma — are genuine; but the tritium inventory and the activated structure are real hazards that must be quantified, not waved away.

THE POSTURE IN ONE LINE

A **minimized mobilizable-tritium source term, no criticality or meltdown mechanism by construction**, and a **low-activation-waste target** — with the governing quantity, the standing tritium inventory, *computed, not asserted*.

§ 2

The dominant hazard: mobilizable tritium

The accident source term is set by the mobilizable tritium inventory. The design minimizes it through direct internal recycling of unburned fuel, which avoids a large buffered store, and through permeation control at the blanket-coolant boundary and detritiation of process streams. The standing-inventory value is computed in the fuel-cycle model and is the governing safety quantity; we carry it rather than assert a figure, and we flag it as the load-bearing input to the source term. **COMPUTED (FUEL-CYCLE)**

§ 3

No meltdown, by construction

There is no fission-product decay-heat inventory and no chain reaction. The fusion plasma requires several simultaneous conditions — confinement, heating, fuelling, field — and the loss of any one quenches it on a short timescale. An unmitigated loss of control therefore *terminates* the reaction rather than escalating it; there is no physical mechanism for a runaway or a core melt. The decay heat that remains is from activated structure, not fuel, and is bounded by the material selection.

Waste, activation, and the safety basis

Structural activation is minimized by low-activation steel selection, targeting a low-level-waste classification of the retired structure and, where practical, short-cooldown or hands-on maintenance. The consumable centrepost cartridge is designed for scheduled handling within this waste class, and the burner’s 5.44% neutron fraction — an order of magnitude under deuterium–tritium — further reduces activation at the power plant.

MOBILIZABLE TRITIUM	Minimized by direct internal recycling · <i>computed</i> (fuel-cycle companion)
CRITICALITY / MELTDOWN	None — no fissile, no chain reaction · <i>inherent</i>
DECAY HEAT	Activated structure only, bounded · <i>requirement</i> (materials)
STRUCTURAL WASTE	Low-activation steel, low-level-waste target · <i>requirement</i> (materials)
POWER-PLANT ACTIVATION	Reduced by the burner’s 5.44% neutron fraction · <i>computed</i> (burner)

This is a design-basis safety argument, not a licensed safety analysis report: the source term depends on the computed standing inventory carried from the fuel-cycle companion, and the waste classification is a material-selection target pending qualification. Regulatory treatment is developed in the licensing companion. What the case establishes is that every hazard is *bounded*, and the one that governs is computed.

DESIGN-BASIS

APPARATUS

Portfolio, People & Record

The intellectual-property estate, the people who built it, the limitations that gate it, and the sources it rests on.

1

GRANTED PATENT

4

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INTELLECTUAL PROPERTY

The patent portfolio

The estate spans a granted high-field magnet patent, pending utility and 2026 provisional filings that map to the two products, a digital-twin control provisional, a registered trademark application, and the original 2022 provisional family. Every patentable disclosure in the five-paper arXiv drop was protected **before** publication: the breeder and burner provisionals were filed 1–2 August, and a publication-gap omnibus filing the evening before the drop swept up the DEC, plasma-control, and REBCO-winding matter of the three otherwise-unprotected papers. Patent numbers and application serials are matters of public record; claim scope is summarised, not reproduced.

GRANTED & PENDING UTILITY

US 12,009,112	High-field tilted graded-REBCO magnet architecture (KRONO-002A) · app. 17/878,550	GRANTED
US 17/878,507	Core device / method (KRONO-001A) · utility	PENDING

2026 PROVISIONAL FILINGS — MAPPED TO THE PRODUCTS

64/124,209	Hyperion — spherical-tokamak tritium / helium-3 foundry · breeder · filed Aug 2026	FILED
64/124,220	Aegis / MetroVolt — synchrotron-cutoff, fuel-selection & plug-winding · generator · filed Aug 2026	FILED
64/115,167	KRONOS-CTRL digital-twin plant control · provisional · filed Jul 2026	FILED
64/005,440	Integrated spherical-tokamak fusion architecture · early integrated-architecture filing · filed Mar 2026	FILED
64/105,530	Negative-triangularity spherical-tokamak architecture · foundational ST filing · filed Jul 2026	FILED
64/128,097	Publication-gap omnibus — quasineutral two-species expander DEC · provenance-gated / latency-split plasma control · as-built high-field winding & conductor architecture · filed 7 Aug 2026, the evening before the arXiv drop	FILED

TRADEMARK & ORIGIN

FTK-98801894	Kronos Fusion Energy — trademark application	FILED
KR-2022-★	Original provisional family (KR-2022-00001 ... 00027) · 27 filings, 2022	PRIORITY

The narrow, defensible magnet novelty — the specific conductor and the digital-twin winding optimisation, not high-field REBCO as a category — is the commercial core that can earn ahead of any $Q > 1$ milestone, with markets in fusion magnet supply, MRI/NMR, accelerators, and proton therapy. The claim is scoped honestly: the high field is a *system field* result, not a stand-alone-coil record; the small-bore plug coil is structurally infeasible as a bare winding but resolved by a stress-managed structural shield (feasible-pending-FEA); and the winding-tape experiment returned a *null result*. The value is the method, not a field record.

THE TEAM

Founding partners, board, advisors & auditors

Kronos is built by a bench of advanced-fuel-fusion, high-field-magnet, direct-conversion, and materials specialists, with a board and operations team drawn from defense, national laboratories, and industry. Dates are shown as ranges; where a tenure has ended or is term-ending, the range says so.

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Legal counsel is retained; those roles are held on the internal roster and are not listed here.

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COLOPHON

How this document was made

The Kronos Fleet — Combined Editorial, 2026 is set in **Fraunces** (display), **Gelasio** (text), and **IBM Plex Mono** (data & equations), carried forward from the Kronos editorial design system.

The figures are rendered at 300 dpi from the deposited generator scripts; the document is composed as a single self-contained file with embedded fonts and imagery, paginated to US Letter, and rendered to PDF through a headless Chromium engine in matched light and dark editions.

Every headline quantity carries an evidence class; withdrawn and restated values are marked as such. Nothing herein is frozen beyond the founder-approved Tier-A set.

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Numbers and their classes. Quantities are reported with evidence classes and case labels; modelled, assumed, and requirement-class values are identified as such and are not to be quoted without their labels.

Public edition. This edition carries the physics and design only; all financial, funding, and defense-commercial content has been removed for public distribution. The scientific text corresponds to the arXiv and journal submissions.

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THE 2026 KRONOS SET

One programme, in five papers

This editorial is one of five that together describe the Kronos fusion programme — a breeder that funds a generator, and the magnet, conversion, and control science that enable both. Each is published open-access with a reproducible companion deposit.

PAPER I	Hyperion — The Tritium Foundry · 10.5281/zenodo.21746157
PAPER II	Aegis & MetroVolt — The Generator · 10.5281/zenodo.21746479
PAPER III	High-Field REBCO Magnets & Tape · 10.5281/zenodo.21842514
PAPER IV	Direct Energy Conversion · 10.5281/zenodo.21842864
PAPER V	AI, ML & Quantum Control · 10.5281/zenodo.21842371

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